

Gene Analysis GUI - Manual

1) Installation

- a. Requirements: Matlab R2024b or newer required Image Processing Toolbox, Statistics and Machine Learning Toolbox
- b. Download and if needed unpack GUI code:
SC_GeneBrowser4.3Deployment Lite into a **single dedicated folder**.
- c. Either:
 - Add that folder with all subfolders to the MATLAB path (Home → Set Path → Add with Subfolders...), or
 - Set the current working folder in MATLAB to the folder that the files were unpacked to.
- d. Download data in compatible format:
deFaria_Carballo_Capitan_data.mat save it on hard drive.

2) Launch

- a. Run **GeneBrowser.m** either through command window
run(GeneBrowser.m) or double click **GeneBrowser.m** in working folder.
- b. The GUI initializes and a file dialog will appear prompting for data source - navigate and load data: '**deFaria_Carballo_Maidana_data.mat**'

3) Initialization

- a. Click 'New Figure' to start exploring.
- b. Some gene name should be always inserted in **Gene...** field – advillin is a good default option '**Avil**'
- c. Click 'GO' to update data visualization shown in the figure.

4) Menus& Controls

Exploration (You are exploring...) Choose what's visualized through cells colorcode on the map. Note all options work with every possible cell subset (control or stimuli ensemble):

- **Normalized expression** - log, cell-wise normalized expression of the selected gene.
- **Raw gene counts** - counts from the gene-by-cell matrix.
- **CaMPARI fluorescence levels** - background-corrected fluorescence (AU) for CaMPARI positive cells. Choose **red** or **green** channel.
- **Louvain clusters** - cells colored by assigned cluster (same color code as in the manuscript).
- **Area** - cell area in pixels.

Control fields

- **In...** - filter to subset of cells to visualize: *All*, *Control*, or a specific stimulus ensemble (all downstream analyses respect this filter).

- **Gene...** - select the gene when using *Normalized expression* or *Raw gene counts* browsing. All gene names should be started with uppercase followed by lowercase only.
- **New figure** - keep the current analysis figure and open a new one for further analysis.
- **GO!** - apply changes after modifying inputs (subset/gene/options).

Visualization scaling

- **Autoscale default** - automatic dynamic range. The number excludes the lowest X% of expressors from display and sets top X% expressors at saturation.
- **Manual** - set min/max explicitly or via sliders.

Box plots

- Create box and scatter plots of gene expression across classes (for all currently filtered in cells).
- You can change the stimulus order using standard MATLAB vector syntax (recommended to keep at default).

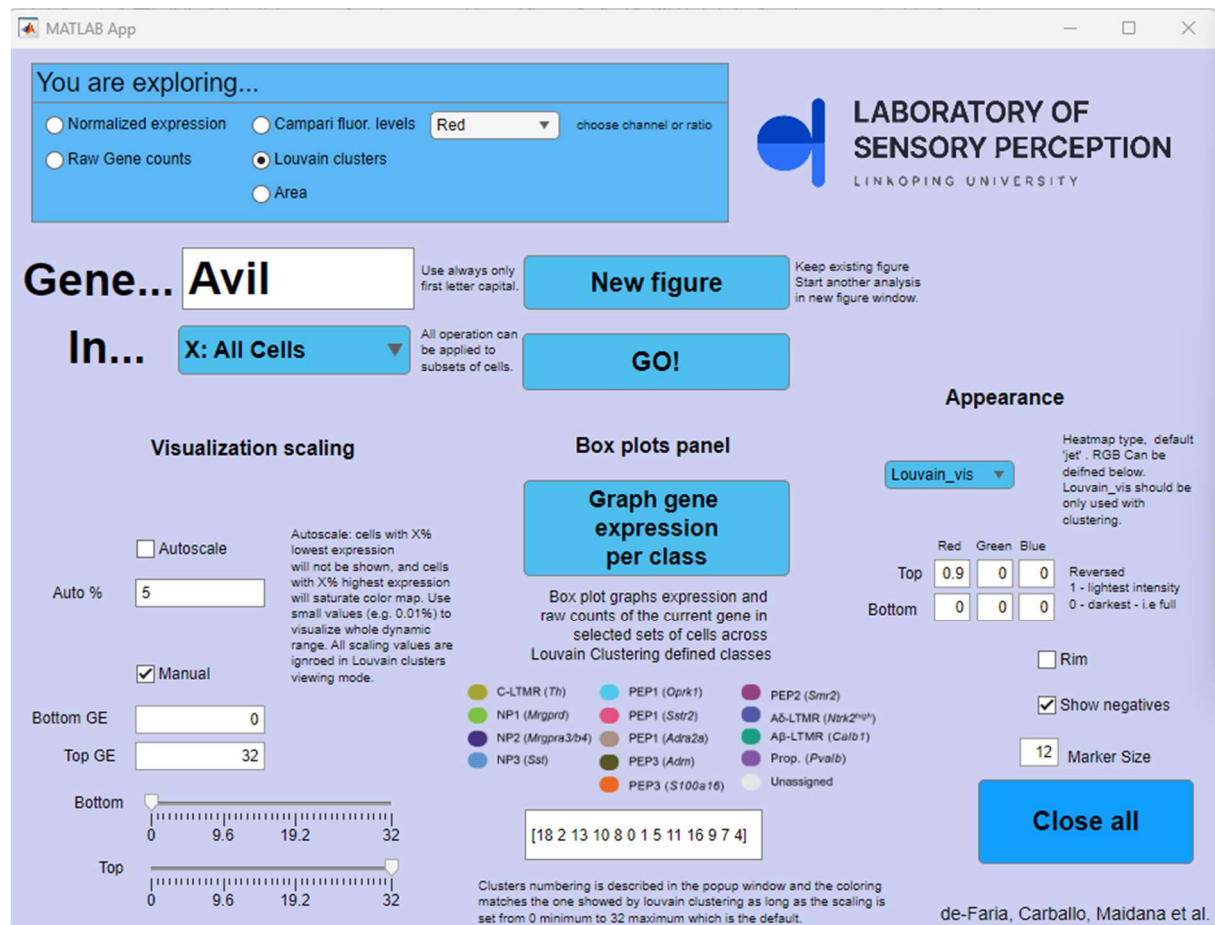
Appearance

- Heatmap styling: choose heatmap type (including custom RGB), draw cells via face or rim coloring, include negatives in grey, and adjust marker size.

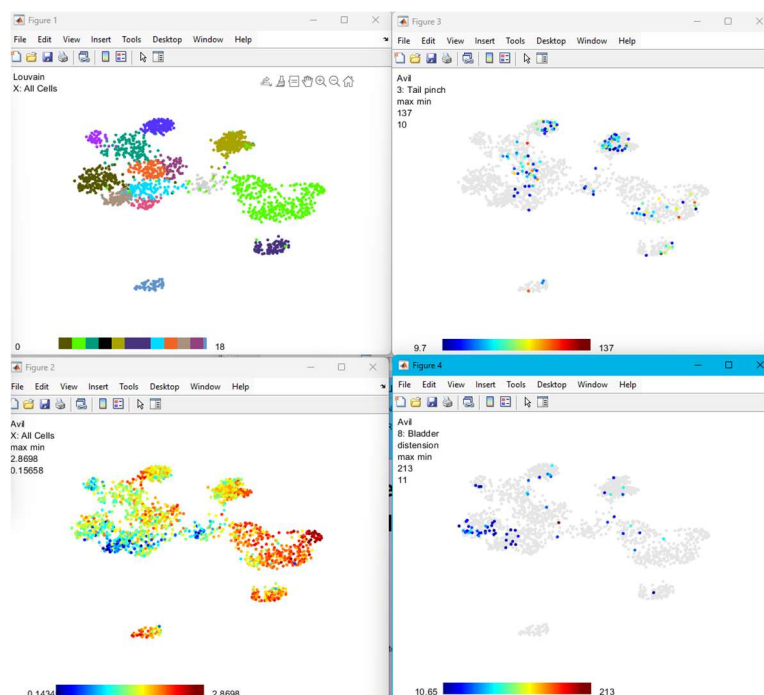
4) Tips

- After changing subset or gene, remember to hit **GO!** to refresh.
- Use New figure to compare different genes/scales side-by-side.
- When using Autoscale, small X% values make rare high expressors more visible; larger values smooth out extremes.

Main window:



UMAP expression analysis windows:



Gene expression analysis windows:

